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Roll No. :

UGM-10032

BCA (Semester-V) Examination, 2025-26

(New Course)

.NET FRAMEWORK AND C#

[Paper : First (BCA 125)]

Time : Three Hours]

[Maximum Marks : 60

Note : Attempt **all** questions. Use code snippets where applicable.

1. (a) Discuss how CLR enables interoperability among multiple .NET languages ? [6]
- (b) Explain how objects interact through method invocation and reference passing? [6]

OR

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(1)

[P.T.O.]

- (a) Explain, how MSIL enables platform independence? [6]
- (b) Discuss, how accessibility modifiers influence inheritance behaviour? Explain using an example. [6]
2. (a) Create an abstract class 'Shape' with abstract method 'Area()'; implement it for circle and rectangle. What happens if it 'Area()' is not implemented by the child classes? [6]
- (b) Describe how delegates encapsulate a method reference and enable callback mechanisms? [6]

OR

- (a) Write a delegate that accepts two integers and returns their product; demonstrate invocation. [6]
- (b) Using an example, discuss how absence of an explicitly defined base class constructor affects derived class construction when only parameterized constructors exist. [6]
3. (a) Create a thread using the 'ThreadStart' delegate in C#. Explain your code and demonstrate the execution. [6]

- (b) Create a program for acquiring and releasing a Mutex in a critical section. Explain your logic as well. [6]

OR

- (a) Demonstrate how to detect thread completion using the 'IsAlive' property in C#. [6]
- (b) Write a program to create 3 threads and control their exit without using any inbuilt methods. [6]
4. (a) Define assemblies in .NET and list its main components. [6]
- (b) Describe when an assembly becomes a shared assembly. [6]

OR

- (a) Discuss the RTTI operators in C# with examples. [6]
- (b) Write a program to count the number of parameters in a method of a class. [6]
5. (a) Define ADO.NET and describe its role in data access. [6]
- (b) Explain, how connection strings work for establishing database connectivity. Use an example to explain your answer. [6]

OR

- (a) Write a C# program that connects to SQL Server using 'SqlConnection' and prints a success message when the connection opens. [6]
- (b) How does ADO.NET make it possible for working with multiple types of databases? List the names of a few classes used for this. [6]

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BCA (Semester-V) Examination, 2025-26

(Old Course)

.NET FRAMEWORK USING C#

[Paper : First (BCA 125)]

Time : Three Hours]

[Maximum Marks : 60

Note : Attempt any five questions. Use code snippets wherever applicable.

1. (a) Enlist the major components of the .Net framework and explain the role of each component. Explain the importance of CLR in it. [6]

- (b) What is Polymorphism? Using a program, explain how compile time and runtime polymorphism is implemented in C#. [6]

2. (a) What are Properties? Why are they needed? How are they different from fields? Explain using an example. [6]
- (b) What are Delegates? What is delegate multicasting and why is it used? What are the restrictions imposed while implementing it? [6]
3. (a) Which are the 3 utility methods present in the Exception base class which can be used for debugging a code. [6]
- (b) What is the difference between a generic catch and a catch-all block? [6]
4. (a) What do you understand by signing an assembly? Explain the need for it. Also explain the role of the different tools used in the process. [6]
- (b) What are the different ways in which one can categorize assemblies? Explain with suitable example. [6]

5. (a) Write a short note on the use of threads. Explain the difference between multitasking and multiprogramming. [6]
- (b) Write a program to create a Mutex. Also discuss its execution. [6]
6. (a) What is the use of structures? Compare it with a class. Explain its working through an example. [6]
- (b) How are enumerations helpful in writing a program? Explain with an example. [6]
7. (a) How are interfaces different from abstract classes? [6]
- (b) Write a program to overload the '&' operator for a class. [6]
8. (a) Discuss the ADO.NET framework. Mention a few important classes used in ADO.NET and their roles in the framework. [6]
- (b) What is the role of a DataSet in ADO.NET?